THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair $x + 3y = 4$ and $3x + ky = 13$ has no solution when $k =$ 1

(a) 12 **(b)** 9 **(c)** 10 **(d)** 3

2. The point where the lines $5x + 3y = 29$ and $4x + 7y = 37$ intersect has $x + y =$ 1

(a) 3 **(b)** 4 **(c)** 7 **(d)** 1

3. The pair of equations $x - 2y = 3$ and $2x + y = 5$ is: 1

(a) exactly one solution (intersecting lines)
(b) no solution (parallel lines)
(c) exactly two solutions
(d) infinitely many solutions (coincident lines)

4. The solution of $x + 2y = 5$, $3x - y = 8$ has $x =$ 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** -3

5. The solution of $4x + 3y = 18$, $3x + 2y = 13$ has $y =$ 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** -2

6. For the pair $x + y = 16$ and $x - y = 6$, the value of xy is: 1

(a) 110 **(b)** 55 **(c)** 73 **(d)** 16

7. The pair $x + 2y = 5$ and $4x + ky = 21$ has no solution when $k =$ 1

(a) 20 **(b)** 2 **(c)** 8 **(d)** 9

8. The point where the lines $6x + 5y = 60$ and $7x + 4y = 59$ intersect has $x + y =$ 1

(a) -1 **(b)** 11 **(c)** 5 **(d)** 6

9. The pair of equations $6x - 3y = 10$ and $2x - y = 9$ is: 1

(a) no solution (parallel lines)
(b) exactly one solution (intersecting lines)
(c) exactly two solutions
(d) infinitely many solutions (coincident lines)

10. The solution of $3x + y = 11$, $x + y = 7$ has $x =$ 1

(a) 2 **(b)** -2 **(c)** 7 **(d)** 5

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by substitution: $2x + y = 10$, $x - y = 2$. 2

12. Find the value of k for which the pair $x + 3y = 4$, $3x + ky = 13$ has no solution. 2

13. A man tells his daughter, "Seven years ago I was 5 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). 2

14. Solve by substitution: $x + 2y = 5$, $3x - y = 8$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On comparing the ratios, find whether the lines $6x - 3y = 10$ and $2x - y = 9$ intersect, are parallel or coincide. 3

16. Solve by elimination method: $4x + 3y = 18$ and $3x + 2y = 13$. 3

17. The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 63 less. Find the number. 3

OR

On comparing the ratios, find whether the lines $3x + 2y = 5$ and $9x + 6y = 15$ intersect, are parallel or coincide. 3

18. Solve by elimination method: $5x + 2y = 20$ and $3x + 4y = 26$. 3

OR

The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 9 less. Find the number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A boat goes 30 km upstream and 45 km downstream in 11 hours, and 40 km upstream and 27 km downstream in 11 hours. Find the speed of the boat in still water and the speed of the stream. 5

OR

If the length of a rectangle is reduced by 5 units and its breadth is increased by 3 units, its area decreases by 30 square units. If the length is increased by 3 units and the breadth by 2 units, the area increases by 91 square units. Find the length and breadth. 5

20. A boat goes 28 km upstream and 30 km downstream in 10 hours, and 21 km upstream and 50 km downstream in $\frac{41}{4}$ hours. Find the speed of the boat in still water and the speed of the stream. 5

OR

If the length of a rectangle is reduced by 4 units and its breadth is increased by 2 units, its area decreases by 20 square units. If the length is increased by 2 units and the breadth by 3 units, the area increases by 84 square units. Find the length and breadth. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND